



Association Analysis

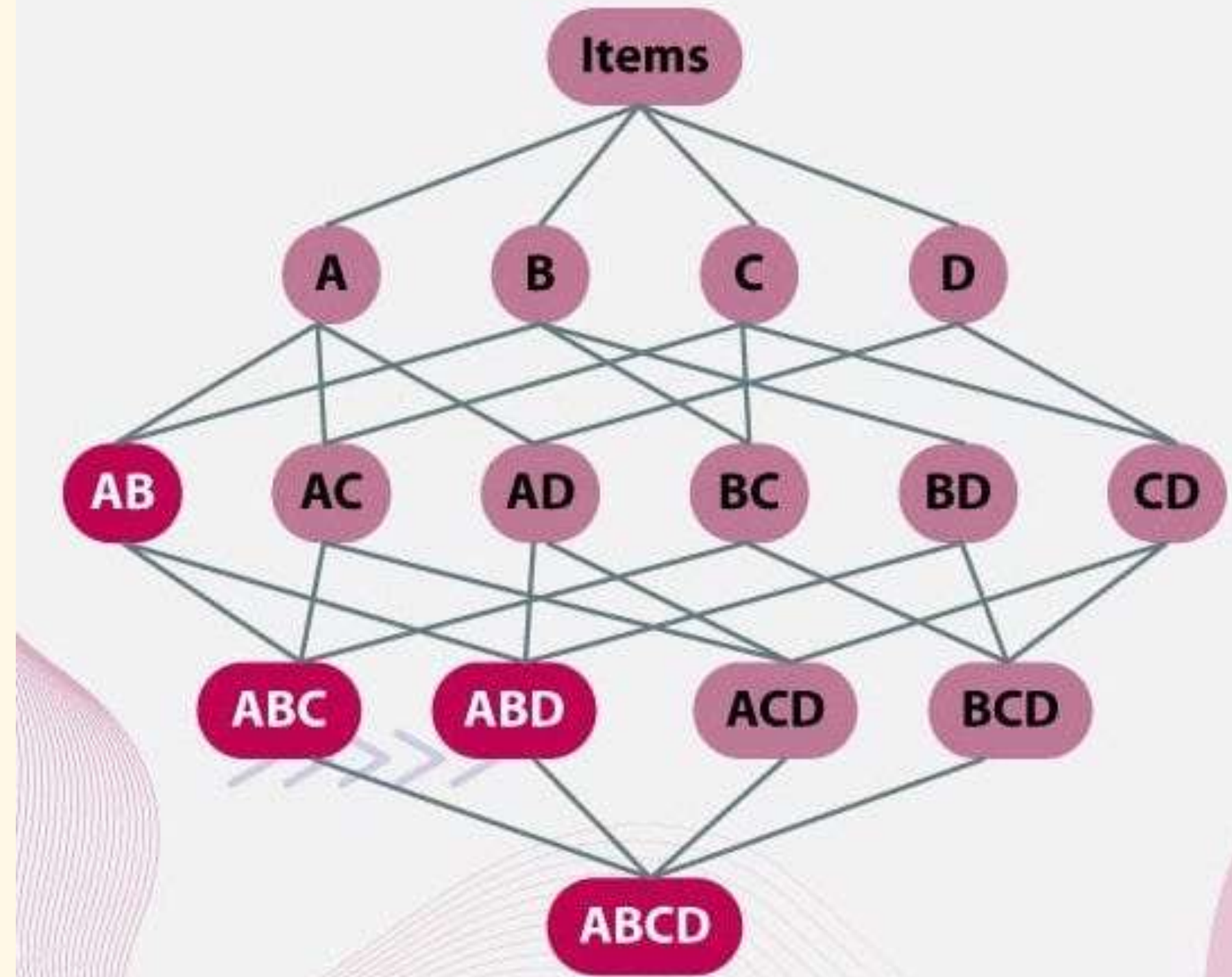
Data Mining

BSCS 3A

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OUTLINE

- ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)
- Mining Frequent Itemsets (APRIORI)
- Concise Itemset Representation
- Alternative Methods to Find Frequent Itemsets
- Pattern Evaluation



PRE-ACTIVITY: “The Missing Combo”

Scenario: You are now the marketing manager of Jollibee. Sales are good—but your boss believes: ‘We are missing opportunities to increase orders per customer.’ So your task is to:

- **Design better combo meals to increase sales.**

“Here are recent customer orders:”

Customer	Orders
1	Chickenjoy, Pie, Coke, Sundae
2	Burger Steak, Coke
3	Chickenjoy, Fries, Coke
4	Chickenjoy, Pie, Fries, Burger
5	Burger Steak, Coke

PRE-ACTIVITY: “The Missing Combo”

“If a customer orders **Chickenjoy**, what else should we recommend?”

- ☐ Sundae
- ☐ Coke
- ☐ Fries
- ☐ Pie
- ☐ Burger

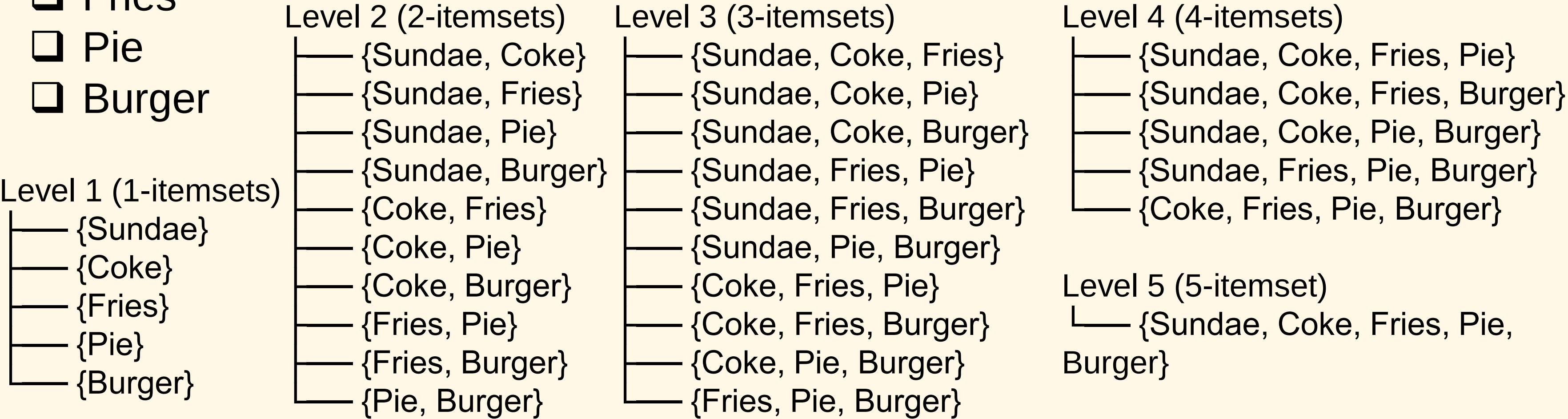
“Which combination appears MOST frequently?”

Customer	Orders
1	Chickenjoy, Pie, Coke, Sundae
2	Burger Steak, Coke
3	Chickenjoy, Fries, Coke
4	Chickenjoy, Pie, Fries, Burger
5	Burger Steak, Coke

PRE-ACTIVITY: “The Missing Combo”

“If a customer orders **Chickenjoy**, what else should we recommend?”

- ☐ Sundae
- ☐ Coke
- ☐ Fries
- ☐ Pie
- ☐ Burger



PRE-ACTIVITY: “The Missing Combo”

“If you can only create ONE combo meal, what would it be?”

“What you just did...
is actually a formal data mining technique
called **Association Analysis.**”

PRE-ACTIVITY: “The Missing Combo”

“Instead of guessing combos, we use data to find:
Which items are frequently bought together”

At Jollibee:

Better combos → higher sales

Better recommendations → better customer experience

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Association Analysis is a data mining technique used to **discover relationships or patterns among items in a dataset.**

It answers questions like:

“What items are frequently bought together?”

“If a customer buys X, what else are they likely to buy?”

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Key Difference from Classification

Aspect	Classification	Association Analysis
Goal	Predict a label	Find relationships
Output	Class (High/Low)	Rules ($X \rightarrow Y$)
Example	High Sales?	Chickenjoy $\rightarrow$ Coke

Association analysis finds **co-occurrence patterns**

✓ If items appear together frequently $\rightarrow$ they are **associated**

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

At Jollibee:

- Customers who buy **Burger Steak** also buy **Coke**
- Customers who buy **Chickenjoy** often buy **Fries**
- ✓ These are **hidden patterns in data**

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

KEY TERMINOLOGIES

1. Itemset

- A collection of items

Example:

{Chickenjoy, Coke}

2. Transaction

- A single purchase

Example:

T1 = {Chickenjoy, Pie, Coke, Sundae}

3. Frequent Itemset

- An itemset that appears frequently

4. Association Rule

- A rule of the form: $x \rightarrow y$

Example:

{Burger Steak} $\rightarrow$ {Coke}

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

STEP-BY-STEP EXAMPLE

Step 1: Count Frequency of Items

Item (L1)	Count	Support
Coke	4	0.80
Chickenjoy	3	0.60
Burger Steak	2	0.40
Pie	2	0.40
Fries	2	0.40
Sundae	1	0.20
Burger	1	0.20

Support = (Number of transactions containing item) / (Total transactions)

Support tells you:

How often an item or itemset appears in the entire dataset

Step 2: Identify Frequent Items

If:

- ✓ Minimum support = 1 (count)

Frequent items:

- Chickenjoy, Coke, Burger Steak, Pie, Fries , Sundae,

Step 3: Find Item Combinations

Itemset (L2)	Count	Support
Chickenjoy, Pie	2	0.40
Chickenjoy, Coke	2	0.40
Chickenjoy, Sundae	1	0.20
Chickenjoy, Fries	2	0.40
Chickenjoy, Burger	1	0.20
Pie, Coke	1	0.20
Pie, Sundae	1	0.20
Pie, Fries	1	0.20
Pie, Burger	1	0.20
Coke, Sundae	1	0.20
Coke, Fries	1	0.20
Burger Steak, Coke	2	0.40

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Step 4: Add Confidence (Both Directions)

1. Chickenjoy, Pie (2/5)

Chickenjoy $\rightarrow$ Pie = $2/3 = 0.67$

Pie $\rightarrow$ Chickenjoy = $2/2 = 1.00$

2. Chickenjoy, Coke (2/5)

Chickenjoy $\rightarrow$ Coke = $2/3 = 0.67$

Coke $\rightarrow$ Chickenjoy = $2/4 = 0.50$

3. Chickenjoy, Sundae (1/5)

Chickenjoy $\rightarrow$ Sundae = $1/3 = 0.33$

Sundae $\rightarrow$ Chickenjoy = $1/1 = 1.00$

4. Chickenjoy, Fries (2/5)

Chickenjoy $\rightarrow$ Fries = $2/3 = 0.67$

Fries $\rightarrow$ Chickenjoy = $2/2 = 1.00$

5. Chickenjoy, Burger (1/5)

Chickenjoy $\rightarrow$ Burger = $1/3 = 0.33$

Burger $\rightarrow$ Chickenjoy = $1/1 = 1.00$

6. Pie, Coke (1/5)

Pie $\rightarrow$ Coke = $1/2 = 0.50$

Coke $\rightarrow$ Pie = $1/4 = 0.25$

7. Pie, Sundae (1/5)

Pie $\rightarrow$ Sundae = $1/2 = 0.50$

Sundae $\rightarrow$ Pie = $1/1 = 1.00$

8. Pie, Fries (1/5)

Pie $\rightarrow$ Fries = $1/2 = 0.50$

Fries $\rightarrow$ Pie = $1/2 = 0.50$

9. Pie, Burger (1/5)

Pie $\rightarrow$ Burger = $1/2 = 0.50$

Burger $\rightarrow$ Pie = $1/1 = 1.00$

10. Coke, Sundae (1/5)

Coke $\rightarrow$ Sundae = $1/4 = 0.25$

Sundae $\rightarrow$ Coke = $1/1 = 1.00$

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

11. Coke, Fries (1/5)

Coke $\rightarrow$ Fries = $1/4 = 0.25$

Fries $\rightarrow$ Coke = $1/2 = 0.50$

12. Burger Steak, Coke (2/5)

Burger Steak $\rightarrow$ Coke = $2/2 = 1.00$

Coke $\rightarrow$ Burger Steak = $2/4 = 0.50$

INTERPRETATION

Strong Rules (Confidence = 1.00)

Moderate Rules (Confidence = 0.50 – 0.67)

Weak Rules (low usefulness) = 0.25

Direction matters a lot

Some rules look strong only one-way (e.g., Sundae $\rightarrow$ Pie = 1.00)

But are weak the other way around

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Step 5: Add Lift Formula

$$\text{Lift}(A \rightarrow B) = \frac{P(A \cap B)}{P(A) \cdot P(B)}$$

Equivalent form:

$$\text{Lift}(A \rightarrow B) = \frac{\text{Confidence}(A \rightarrow B)}{\text{Support}(B)}$$

Item	Support
Chickenjoy	3/5 = 0.60
Pie	2/5 = 0.40
Coke	4/5 = 0.80
Fries	2/5 = 0.40
Burger Steak	2/5 = 0.40
Sundae	1/5 = 0.20
Burger	1/5 = 0.20

- **Numerator:** $P(A \cap B)$
→ Probability that **A** and **B** occur together
- **Denominator:** $P(A) \cdot P(B)$
→ Expected probability if **A** and **B** were independent

Rule	Lift
Chickenjoy → Pie	(0.67 / 0.40) = 1.67
Chickenjoy → Coke	(0.67 / 0.80) = 0.83
Chickenjoy → Fries	(0.67 / 0.40) = 1.67
Chickenjoy → Burger Steak	(0.00 / 0.40) = 0.00
Chickenjoy → Sundae	(0.33 / 0.20) = 1.67
Chickenjoy → Burger	(0.33 / 0.20) = 1.67
Pie → Chickenjoy	(1.00 / 0.60) = 1.67
Pie → Coke	(0.50 / 0.80) = 0.63
Pie → Fries	(0.50 / 0.40) = 1.25
Pie → Burger Steak	(0.00 / 0.40) = 0.00
Pie → Sundae	(0.50 / 0.20) = 2.50
Pie → Burger	(0.50 / 0.20) = 2.50

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Step 5: Add Lift Formula

$$\text{Lift}(A \rightarrow B) = \frac{P(A \cap B)}{P(A) \cdot P(B)}$$

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Item	Support
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Pie	2/5 = 0.40
Coke	4/5 = 0.80
Fries	2/5 = 0.40
Burger Steak	2/5 = 0.40
Sundae	1/5 = 0.20
Burger	1/5 = 0.20

Rule	Lift
Coke → Chickenjoy	(0.50 / 0.60) = 0.83
Coke → Pie	(0.25 / 0.40) = 0.63
Coke → Fries	(0.25 / 0.40) = 0.63
Coke → Burger Steak	(0.50 / 0.40) = 1.25
Coke → Sundae	(0.25 / 0.20) = 1.25
Coke → Burger	(0.00 / 0.20) = 0.00
Fries → Chickenjoy	(1.00 / 0.60) = 1.67
Fries → Pie	(0.50 / 0.40) = 1.25
Fries → Coke	(0.50 / 0.80) = 0.63
Fries → Burger Steak	(0.00 / 0.40) = 0.00
Fries → Sundae	(0.00 / 0.20) = 0.00
Fries → Burger	(0.50 / 0.20) = 2.50

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Step 5: Add Lift Formula

$$\text{Lift}(A \rightarrow B) = \frac{P(A \cap B)}{P(A) \cdot P(B)}$$

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Item	Support
Chickenjoy	3/5 = 0.60
Pie	2/5 = 0.40
Coke	4/5 = 0.80
Fries	2/5 = 0.40
Burger Steak	2/5 = 0.40
Sundae	1/5 = 0.20
Burger	1/5 = 0.20

Rule	Lift
Burger Steak → Coke	(1.00 / 0.80) = 1.25
Sundae → Chickenjoy	(1.00 / 0.60) = 1.67
Sundae → Pie	(1.00 / 0.40) = 2.50
Sundae → Coke	(1.00 / 0.80) = 1.25
Burger → Chickenjoy	(1.00 / 0.60) = 1.67
Burger → Pie	(1.00 / 0.40) = 2.50
Burger → Fries	(1.00 / 0.40) = 2.50

Neutral Association (≈1)
Burger Steak → Coke (**1.25**)
Coke → Burger Steak (**1.25**)
Sundae → Coke (**1.25**)

Weak / Negative Association (<1)
Coke → Pie (**0.63**)
Coke → Fries (**0.63**)
Chickenjoy → Coke (**0.83**)
Coke → Chickenjoy (**0.83**)

INTERPRETATION

Strong Positive Association (Lift > 1.5)
Pie ↔ Sundae (**2.50**)
Pie ↔ Burger (**2.50**)
Fries → Burger (**2.50**)
Sundae → Pie (**2.50**)
Burger → Pie / Fries (**2.50**)
Chickenjoy → Pie / Fries / Sundae / Burger (~**1.67**)

What Lift fixed:
✗ Before (Confidence only)
•Sundae → Chickenjoy = 1.00 (looked strong but misleading)
✓ After (Lift)
•Sundae → Chickenjoy = **1.67 (moderate, not extreme)**
shows it is NOT perfect dependency

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

STEP-BY-STEP EXAMPLE

Step 1: Count Frequency of Items

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Coke	4	0.80
Chickenjoy	3	0.60
Burger Steak	2	0.40
Pie	2	0.40
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Sundae	1	0.20
Burger	1	0.20

Support = (Number of transactions containing item) / (Total transactions)

Support tells you:
How often an item or itemset appears in the entire dataset

Step 2: Identify Frequent Items

If:

✓ **Minimum support = 2 (count)**

Frequent items:

- Chickenjoy, Coke, Burger Steak, Pie, Fries

Step 3: Find Item Combinations

Itemset	Count	Support
Chickenjoy, Pie	2	0.40
Chickenjoy, Coke	2	0.40
Chickenjoy, Sundae	1	0.20
Chickenjoy, Fries	2	0.40
Chickenjoy, Burger	1	0.20
Pie, Coke	1	0.20
Pie, Sundae	1	0.20
Pie, Fries	1	0.20
Pie, Burger	1	0.20
Coke, Sundae	1	0.20
Coke, Fries	1	0.20
Burger Steak, Coke	2	0.40

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Step 4: Add Confidence (Both Directions)

Rule	Confidence
Chickenjoy → Pie	2/3 = 0.67
Pie → Chickenjoy	2/2 = 1.00
Chickenjoy → Coke	2/3 = 0.67
Coke → Chickenjoy	2/4 = 0.50
Chickenjoy → Fries	2/3 = 0.67
Fries → Chickenjoy	2/2 = 1.00
Burger Steak → Coke	2/2 = 1.00
Coke → Burger Steak	2/4 = 0.50

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

INTERPRETATION

Strong Rules (Confidence = 1.00)
Moderate Rules (Confidence = 0.50 – 0.67)
Weak Rules (low usefulness) = 0.25

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

Step 5: Add Lift Formula

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Rule	Lift
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Chickenjoy → Fries	(0.67 / 0.40) = 1.67
Pie → Chickenjoy	(1.00 / 0.60) = 1.67
Coke → Chickenjoy	(0.50 / 0.60) = 0.83
Coke → Burger Steak	(0.50 / 0.40) = 1.25
Fries → Chickenjoy	(1.00 / 0.60) = 1.67
Burger Steak → Coke	(1.00 / 0.80) = 1.25

STRONG ASSOCIATIONS (Lift ≥ 1.5)

Chickenjoy ↔ Pie (**1.67**)

Chickenjoy ↔ Fries (**1.67**)

MODERATE ASSOCIATIONS

Burger Steak ↔ Coke (**1.25**)

Chickenjoy ↔ Coke (**0.83** → **weak negative association**)

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

3 Itemset

Itemset	Count	Support
Chickenjoy, Pie	2	0.40
Chickenjoy, Coke	2	0.40
Chickenjoy, Fries	2	0.40
Burger Steak, Coke	2	0.40

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

Combine 2 Itemsets	Candidate 3-itemset	Valid?	Support Count
(Chickenjoy, Pie) + (Chickenjoy, Coke)	Chickenjoy, Pie, Coke	✗ (Pie—Coke not frequent in L2)	1
(Chickenjoy, Pie) + (Chickenjoy, Fries)	Chickenjoy, Pie, Fries	✗ (Pie—Fries not frequent in L2)	1
(Chickenjoy, Coke) + (Chickenjoy, Fries)	Chickenjoy, Coke, Fries	✗ (Coke—Fries not frequent in L2)	1
(Chickenjoy, Coke) + (Burger Steak, Coke)	Chickenjoy, Coke, Burger Steak	✗ (Chickenjoy—Burger Steak not frequent in L2)	0
(Chickenjoy, Fries) + (Burger Steak, Coke)	✗ cannot join	—	—
(Chickenjoy, Pie) + (Burger Steak, Coke)	✗ cannot join	—	—

There is no 3 Itemset

ASSOCIATION ANALYSIS (MARKET BASKET ANALYSIS)

WHY ASSOCIATION ANALYSIS IS IMPORTANT

Business Value

At Jollibee:

- ✓ Create combo meals
- ✓ Recommend products
- ✓ Increase sales
- ✓ Improve customer experience

Real Examples

- ✓ “People who buy burgers also buy fries”
- ✓ “Customers who buy milk also buy bread”

“Association analysis helps us understand not just what customers buy—but what they buy together, revealing opportunities for smarter decisions.”

APRIORI ALGORITHM (MINING FREQUENT ITEMSETS)

Apriori is an algorithm used to find **frequent itemsets** in transactional data using a level-wise (bottom-up) approach.

“If an itemset is frequent, all of its subsets must also be frequent.” ->

This is called the Apriori Principle

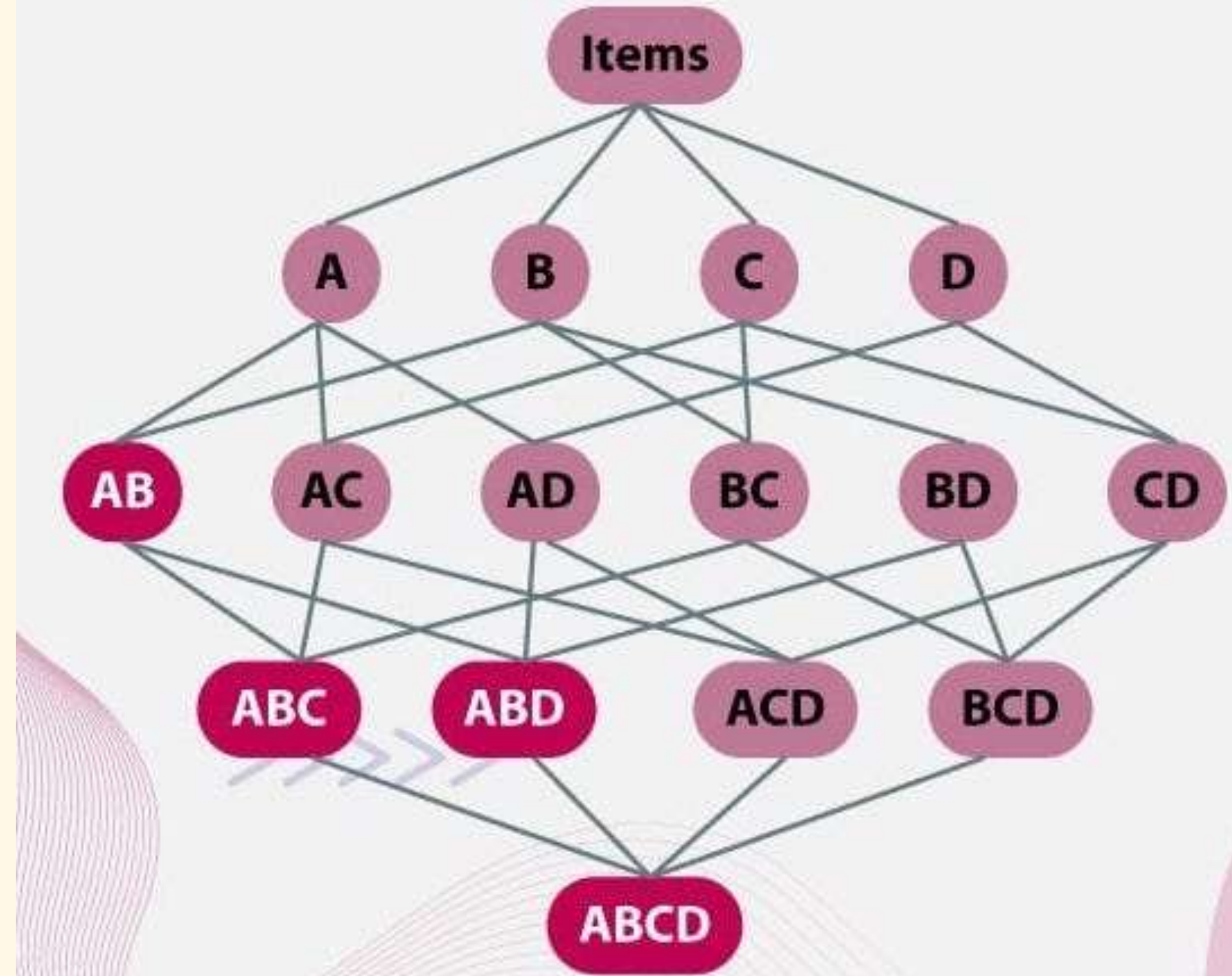
WHY DO WE NEED APRIORI?

Without Apriori:

- ✓ We check ALL possible combinations
- ✓ Too slow (combinatorial explosion)

With Apriori:

- ✓ Eliminate impossible candidates early
- ✓ Efficient pattern discovery



APRIORI-DATASET (JOLLIBEE)

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

SET MINIMUM SUPPORT

Let:

$\text{Min Support} = 2$

- An itemset must appear **at least 2 times**

STEP 1: FIND FREQUENT 1-ITEMSETS (L1)

Item	Count
Chickenjoy	3
Coke	3
Burger Steak	2
Pie	2
Fries	2
Sundae	1 ✕
Burger	1 ✕

Keep Frequent

L1: {Chickenjoy, Coke, Burger Steak, Pie, Fries}

APRIORI-DATASET (JOLLIBEE)

STEP 2: GENERATE 2-ITEMSETS (C2)

Combine L1 items

Candidates:

Chickenjoy + Coke

Chickenjoy + Pie

Chickenjoy + Fries

Chickenjoy + Burger Steak

Coke + Burger Steak

Coke + Pie

Coke + Fries

Pie + Fries

Count Support

Itemset	Count
Chickenjoy + Coke	2
Chickenjoy + Pie	2
Chickenjoy + Fries	2
Coke + Burger Steak	2
Others	< 2 ✕

Keep Frequent (L2)

{(Chickenjoy, Coke), (Chickenjoy, Pie),
(Chickenjoy, Fries), (Burger Steak, Coke)}

APRIORI-DATASET (JOLLIBEE)

STEP 3: GENERATE 3-ITEMSETS (C3) **Apply APRIORI PRUNING**

Possible:

Chickenjoy + Coke + Pie

Chickenjoy + Coke + Fries

Chickenjoy + Pie + Fries

Itemset	Count
Chickenjoy + Coke	2
Chickenjoy + Pie	2
Chickenjoy + Fries	2
Coke + Burger Steak	2
Others	< 2 ✗

Check subsets:

Example:

{CJ, Coke, Pie}

Subsets:

- CJ+Coke ✓
- CJ+Pie ✓
- Coke+Pie ✗

{CJ, Coke, Fries}

Subsets:

- CJ+Coke ✓
- CJ+Fries ✓
- Coke+Fries ✗

•{CJ, Pie, Fries}

Subsets:

- CJ+Pie ✓
- CJ+Fries ✓
- Pie+Fries ✗

“A candidate survives ONLY if ALL its subsets are frequent.”

If no 3-itemsets survive, then your final result will only include frequent 1-itemsets and 2-itemsets.

“Apriori does not just generate combinations—it aggressively eliminates them using prior knowledge.”

APRIORI-DATASET (JOLLIBEE)

What does this mean?

At Jollibee:

Customers commonly buy items in **pairs**,
but not consistently in **triples**.

Business Insight

✓ Strong pairings:

- Chickenjoy + Coke
- Burger Steak + Coke

✗ No strong 3-item combo:

- No consistent “triple meal pattern”

“Why do you think no 3-itemset survived?”

“Apriori stops when no further frequent itemsets can be generated.”

APRIORI-DATASET (JOLLIBEE)

TID	Items
T1	Chickenjoy, Coke, Fries
T2	Chickenjoy, Coke
T3	Chickenjoy, Fries
T4	Coke, Fries
T5	Chickenjoy, Coke, Fries
T6	Chickenjoy, Burger
T7	Coke, Sundae

APPLY APRIORI (Min Support = 2)

STEP 1: L1

Item	Count
Chickenjoy	5
Coke	5
Fries	4
Burger	1 ✗
Sundae	1 ✗

L1: Chickenjoy, Coke, Fries

STEP 3: C3

Candidate: {CJ, Coke, Fries}

Check subsets

CJ + Coke ✓

CJ + Fries ✓

Coke + Fries ✓

Count support

Appears in:

T1, T5 → Count = 2



FINAL L3: {CJ, Coke, Fries}

STEP 2: L2

Itemset	Count
CJ + Coke	3
CJ + Fries	3
Coke + Fries	3

All frequent ✓

CONCISE ITEMSET REPRESENTATION

WHY DO WE NEED IT? is the next level after Apriori.

Problem

After Apriori, we may get:

- ✓ Many frequent itemsets
- ✓ Redundant patterns
- ✓ Hard to interpret results

Example Problem

From your dataset:

- {Chickenjoy} → frequent
- {Chickenjoy, Coke} → frequent
- {Chickenjoy, Coke, Fries} → frequent
- These are **related and overlapping**

“Do we really need to keep ALL of them?”

CONCISE ITEMSET REPRESENTATION

We will use your dataset with:
Minimum support = 2,
STEP 1: Frequent Itemset (Summary from Apriori)

L1 (Frequent 1-itemsets)

Item	Support Count
Chickenjoy	3
Coke	4
Pie	2
Fries	2
Burger Steak	2

L2 (Frequent 2-itemsets)

Itemset	Support Count
Chickenjoy, Pie	2
Chickenjoy, Coke	2
Chickenjoy, Fries	2
Burger Steak, Coke	2

L3 None

CONCISE ITEMSET REPRESENTATION

STEP 2: Concise Representation Types

We now classify itemsets into:

1. CLOSED ITEMSETS

An itemset is **closed** if NO superset has the same support.

Itemset	Support	Why Closed?
Chickenjoy	3	No superset has support 3
Coke	4	No superset has support 4
Pie	2	!Not Closed
Fries	2	!Not Closed
Burger Steak	2	!Not Closed

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

A **superset of Chickenjoy** means:
Any itemset that contains Chickenjoy + additional items

“No superset has support 3” means: No larger combination containing Chickenjoy appears in all 3 of its transactions.

Closed = “already the largest group that shares exactly the same transactions”

CONCISE ITEMSET REPRESENTATION

Itemset	Transactions	Support
Coke	T1, T2, T3, T5	4
Coke + Chickenjoy	T1, T3	2
Coke + Pie	T1	1
Coke + Fries	T3	1
Coke + Burger Steak	T2, T5	2
Coke + Sundae	T1	1
Coke + Burger	—	0

Itemset	Transactions	Support
Pie	T1, T4	2
Pie + Chickenjoy	T1, T4	2 !Not Closed
Fries	T3, T4	2
Fries + Chickenjoy	T3, T4	2 !Not Closed
Burger Steak	T2, T5	2
Burger Steak + Coke	T2, T5	2 !Not Closed

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

If an item always co-occurs with another item in exactly the same transactions, it is NOT closed.

CONCISE ITEMSET REPRESENTATION

Take a 2-itemset, for example:
(Chickenjoy, Pie) with support = 2
This appears in: T1 , T4

Now check its supersets (3-itemsets)

A superset means: Add one more item

Possible supersets:

- (Chickenjoy, Pie, Coke)
- (Chickenjoy, Pie, Fries)
- (Chickenjoy, Pie, Burger)
- (Chickenjoy, Pie, Sundae)

Think of it like this:

- (Chickenjoy, Pie) → customers in T1 and T4
- Ask: “Do BOTH of these customers also share another item?”

Answer: No (T1 and T4 have different extra items)

So: You cannot extend it without losing transactions

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

Check their supports

Itemset	Appears in	Support
Chickenjoy, Pie	T1, T4	2
Chickenjoy, Pie, Coke	T1	1
Chickenjoy, Pie, Fries	T4	1
Chickenjoy, Pie, Burger	T4	1

All supersets have support: **1**, not **2**

There is **no larger itemset** that: contains (Chickenjoy, Pie)

“**No superset with the same support**” means: You cannot add more items to this itemset without reducing its support

CONCISE ITEMSET REPRESENTATION

STEP 2: Concise Representation Types

We now classify itemset into:

2. MAXIMAL ITEMSETS

An itemset is **maximal** if NONE of its supersets are frequent.

We check L2 itemsets:

Itemset	Why Maximal?
Chickenjoy, Pie	No frequent 3-itemset extends it
Chickenjoy, Coke	No frequent 3-itemset extends it
Chickenjoy, Fries	No frequent 3-itemset extends it
Burger Steak, Coke	No frequent 3-itemset extends it

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

Chickenjoy, Pie)

Try extending:

- Chickenjoy, Pie, Coke → support = 1
- Chickenjoy, Pie, Fries → support = 1
- Chickenjoy, Coke, Fries → support = 1

CONCISE ITEMSET REPRESENTATION

STEP 2: Concise Representation Types

We now classify itemsets into:

3. GENERATORS (MINIMAL ITEMSETS)

A generator is an itemset with **no subset having the same support**
It is the **smallest combination that uniquely identifies a pattern**.
So: If you remove any item → support becomes different (usually larger)
So it is “minimal but meaningful”

STEP 1: Check L1 generators

For single items: No subset exists (empty set ignored)
➤ So ALL L1 are generators

Item	Generator?
Chickenjoy	✓
Coke	✓
Pie	✓
Fries	✓
Burger Steak	✓

CONCISE ITEMSET REPRESENTATION

STEP 2: Concise Representation Types

We now classify itemsets into:

3. GENERATORS (MINIMAL ITEMSETS)

A generator is an itemset with **no subset having the same support**
It is the **smallest combination that uniquely identifies a pattern.**

So: If you remove any item → support becomes different (usually larger)
So it is “minimal but meaningful”

STEP 2: Check L2 generators

We compare each L2 itemset with its subsets.

(Chickenjoy, Pie) = 2

Subsets:

- Chickenjoy = 3
- Pie = 2 ! SAME SUPPORT
- violates generator rule
- ✗ NOT a generator

(Chickenjoy, Coke) = 2

Subsets:

- Chickenjoy = 3
- Coke = 4
- none equals 2
- ✓ GENERATOR

(Chickenjoy, Fries) = 2

Subsets:

- Chickenjoy = 3
- Fries = 2 ! SAME SUPPORT
- ✗ NOT generator

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

(Burger Steak, Coke) = 2

Subsets:

- Burger Steak = 2 ! SAME SUPPORT
- Coke = 4
- ✗ NOT generator

CONCISE ITEMSET REPRESENTATION

STEP 2: Concise Representation Types

We now classify itemsets into:

3. GENERATORS (MINIMAL ITEMSETS)

FINAL GENERATOR SET

✓ L1 Generators

- Chickenjoy
- Coke
- Pie
- Fries
- Burger Steak

✓ L2 Generators

- (Chickenjoy, Coke)

Generators = “minimal defining patterns”

They answer:

“What is the smallest itemset that still has this exact support?”

CONCISE ITEMSET REPRESENTATION

STEP 3: FINAL CONCISE SUMMARY

CLOSED ITEMSETS

(Complete but non-redundant representation)

- 2 L1 items
- All L2 frequent pairs above
- ✓ Total: 6 closed itemsets

MAXIMAL ITEMSETS

(Only boundary patterns)

- Chickenjoy, Pie
- Chickenjoy, Coke
- Chickenjoy, Fries
- Burger Steak, Coke
- ✓ Total: 4 maximal itemsets

GENERATORS

(Minimal defining patterns)

- Same as L2 frequent itemsets in this case

Relationship:

Type	Meaning
Frequent itemset	All patterns
Closed itemset	Removes redundancy
Maximal itemset	Only “largest patterns”
Generator	Minimal defining patterns

Concise representation reduces redundancy while preserving all meaningful association information from frequent itemsets.

Alternative Methods to Find Frequent Itemsets

1. FP-Growth (Frequent Pattern Growth)

Idea: Avoid candidate generation completely

Instead of generating combinations:

➤ It **compresses transactions into a tree** and mines patterns

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

How it works:

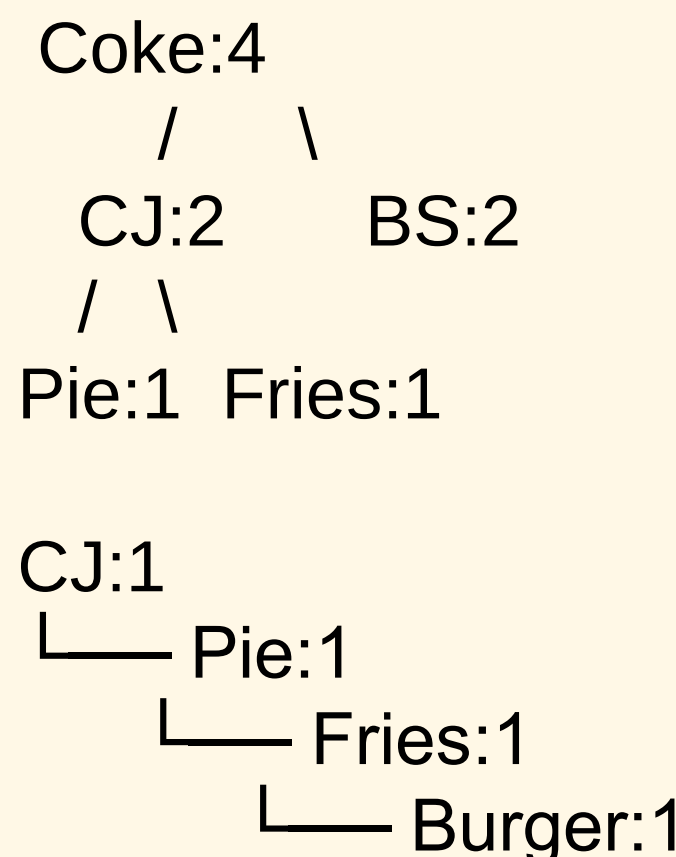
1. Scan database → count item frequencies
2. Build a **compressed tree (FP-tree)**
3. Recursively extract patterns

Advantages:

- ✓ No candidate generation
- ✓ Only **2 database scans**
- ✓ Much faster for large datasets

When to use:

- ✓ Large datasets
- ✓ Dense data (many overlapping items)



Main branch (shared transactions with Coke)

✓ **Coke:4** → appears in 4 transactions

From Coke:

- → **CJ:2**
- → Pie:1
- → Fries:1

• → **BS:2**

Separate branch (no Coke in T4)

✓ **CJ:1** → **Pie:1** → **Fries:1** → **Burger:1**

If **burger is frequent**, then your FP-tree path should NOT stop at Fries.

Alternative Methods to Find Frequent Itemsets

2. ECLAT (Equivalence Class Transformation)

Idea: Use **vertical data format**

Instead of:

Item	Transactions
Chickenjoy	T1, T3, T4

It uses:

Transaction ID sets (TID sets)

Then:

Intersect sets to compute support

Advantages:

- Very fast using set intersections
- Efficient for smaller or medium datasets

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

Example:

- $CJ \rightarrow \{T1, T3, T4\}$
- $Coke \rightarrow \{T1, T2, T3, T5\}$

$CJ \cap Coke = \{T1, T3\} \rightarrow \text{support} = 2$

Why this is useful (ECLAT method)

Instead of scanning the database:

We just do: Set intersection = fast support computation

Pattern Evaluation

Pattern evaluation is the process of **selecting meaningful and useful patterns** from all discovered frequent itemsets or rules.

After mining, you usually get:

! Too many rules

! Many trivial or misleading patterns

➤ So we filter and prioritize **high-quality patterns**

1. Objective Measures (data-driven)

These are **quantitative metrics**.

Support

- Frequency of occurrence
- Helps remove rare patterns
- Already used in mining

Confidence

- Reliability of rule
- But can be misleading

Lift (VERY IMPORTANT)

Measures **true association strength**

- $\text{Lift} > 1 \rightarrow$ positive relationship
- $\text{Lift} = 1 \rightarrow$ independent
- $\text{Lift} < 1 \rightarrow$ negative relationship

Example (from your dataset)

Burger Steak $\rightarrow$ Coke

- Confidence = 1.00
- Lift = 1.25
- ✓ Strong and meaningful

Chickenjoy $\rightarrow$ Coke

Confidence ≈ 0.67

Lift ≈ 0.84

✗ Not strong (below independence)

Pattern Evaluation

2. Other Objective Measures (brief)

- **Leverage** → difference from independence
- **Conviction** → direction strength
- **Correlation** → statistical relationship

Used in advanced analysis/research

3. Subjective Measures (user-driven)

These depend on **domain knowledge**.

4. Pruning Uninteresting Patterns

We remove:

- ✗ Redundant rules
- ✗ Low-lift rules
- ✗ Trivial patterns

Pattern evaluation ensures that only meaningful, non-trivial, and actionable patterns are retained from the mined results.

“From many patterns → keep only the ones that matter.”

Example

We will keep only **strong rules**, meaning:

- ✓ Support ≥ 0.40 (minimum support = $2/5$)
- ✓ Lift > 1 (positive association)
- ✓ Prefer higher confidence

TID	Items
T1	Chickenjoy, Pie, Coke, Sundae
T2	Burger Steak, Coke
T3	Chickenjoy, Fries, Coke
T4	Chickenjoy, Pie, Fries, Burger
T5	Burger Steak, Coke

STEP 1: Candidate Rules (from L2)

From frequent itemsets:

- (Chickenjoy, Coke)
- (Chickenjoy, Pie)
- (Chickenjoy, Fries)
- (Burger Steak, Coke)

STEP 2: Evaluate Rules

We compute:

- Support = count / 5
- Confidence
- Lift

Rule	Support	Confidence	Lift	Interpretation
Burger Steak $\rightarrow$ Coke	0.40	1.00	1.25	Strong: BS buyers always buy Coke
Coke $\rightarrow$ Burger Steak	0.40	0.50	1.25	Coke buyers more likely to buy BS

Example

REMOVED RULES (Why?)

Rule	Reason
Chickenjoy → Coke	Lift < 1 (0.84)
Coke → Chickenjoy	Lift < 1
Chickenjoy → Pie	Lift = 0.83
Pie → Chickenjoy	Lift = 0.83
Chickenjoy → Fries	Lift = 0.83
Fries → Chickenjoy	Lift = 0.83

These are **weak or misleading associations**

KEY INSIGHT

Even though some rules had:

✓ High confidence (e.g., 1.00)

They were rejected because:

✗ Lift ≤ 1 (not truly associated)

FINAL RESULT

✓ Only **Burger Steak ↔ Coke** shows a **true strong relationship**

Business Interpretation

- Bundle: **Burger Steak + Coke**
- Promotion: combo meals
- Inventory: stock together

After evaluation, only Burger Steak–Coke rules show strong, meaningful associations in the dataset.

POST-EXERCISE: Association Rule Evaluation & Filtering

Dataset

TID	Items
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Bread, Butter

Given:
Minimum Support = 2
transactions (0.40)

- Use:
- Support
 - Confidence
 - Lift

Task 1: Find Frequent Itemsets (L1 & L2)

1. Compute support for all **1-itemsets**
2. Compute support for all **2-itemsets**
3. Keep only those with support ≥ 2

Task 2: Generate ALL Possible Rules

From each frequent 2-itemset:

Example: (Milk, Bread) $\rightarrow$

- Milk $\rightarrow$ Bread
- Bread $\rightarrow$ Milk

Task 3: Compute Metrics

For each rule, compute:

- Support
- Confidence
- Lift

Task 4: FILTER STRONG RULES

Keep only rules that satisfy:

- ✓ Support ≥ 0.40
- ✓ Confidence ≥ 0.60
- ✓ Lift > 1



Rule	Support	Confidence	Lift	Interpretation
?	?	?	?	?

POST-EXERCISE:

- Do Machine Exercise: Apriori_Sy.ipynb
- Convert your post-exercise (milk, bread, butter, etc) to this Apriori coding

Thank You!

Any Questions?